

A Head Taller: A Meta-Analysis on the Relation between Pretend Play and Executive Functions in Early Childhood

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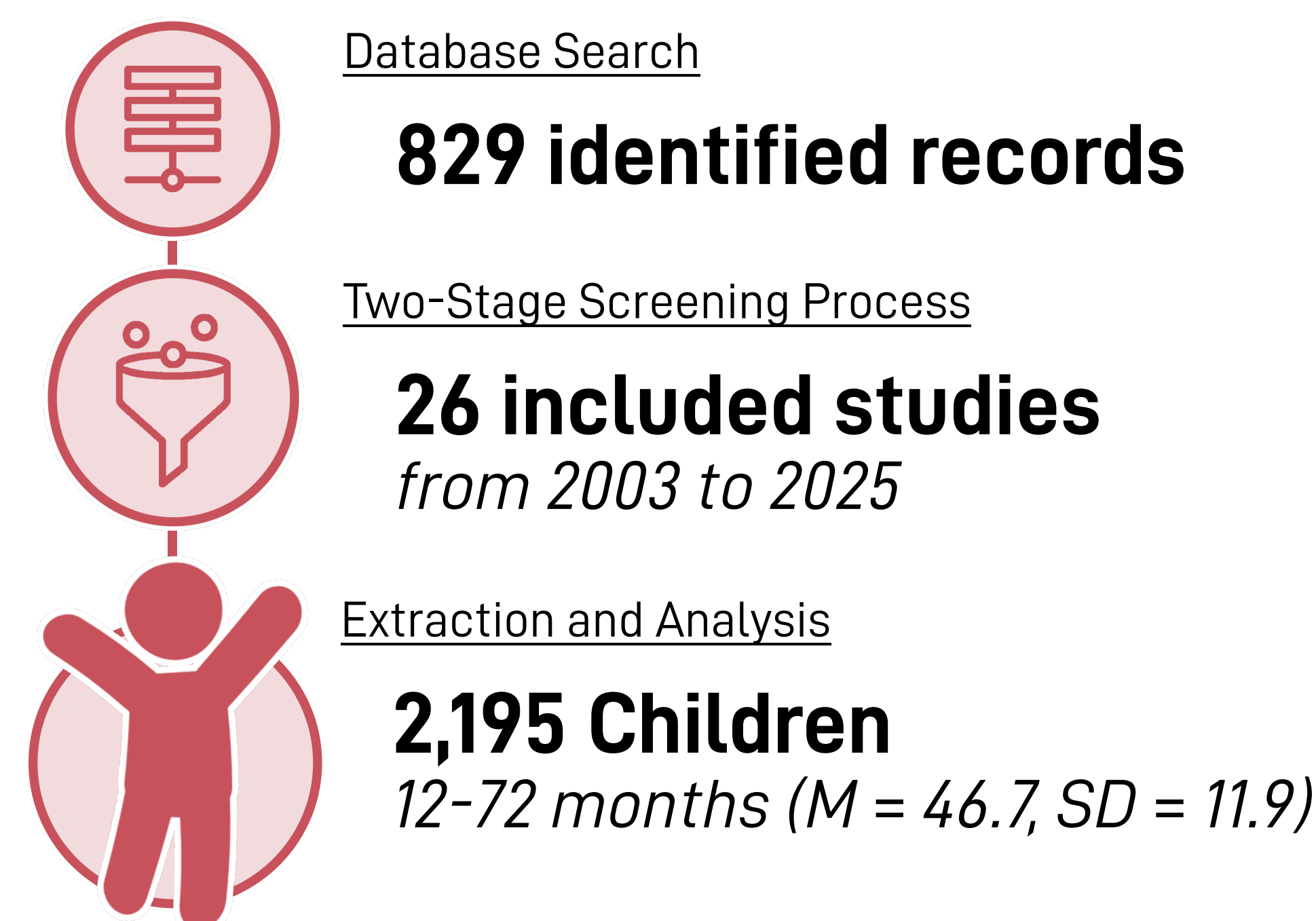
BACKGROUND & OBJECTIVES

- Executive functions** are a set of higher-order cognitive processes that allow control of thoughts, actions, and emotions.
Morgan et al., 2024; Zelazo & Carlson, 2012
- Pretend play**, which engages imagination and creativity in play, has been posited to benefit children's executive development.
Gleason & White, 2023; Vygotsky, 1967
- Yet, contrary to theoretical perspectives, the **overall empirical evidence base** on their relation is not consistent.
Doebel & Lillard, 2023; Lillard et al., 2013

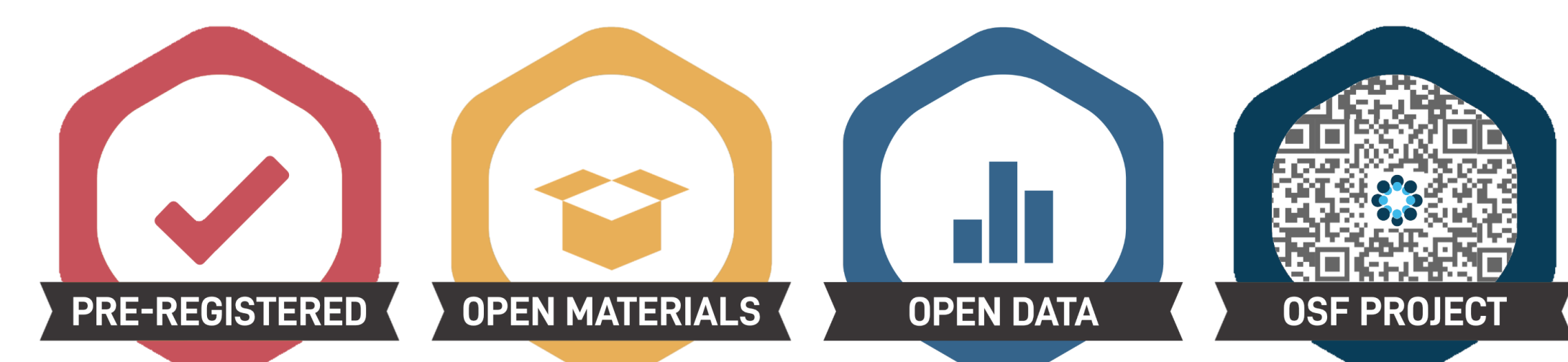
The aim of this study was to identify a quantitative estimate of the relation between pretend play and executive functions in early childhood.

METHODOLOGY

Following a **pre-registered protocol**, five databases were searched for studies identifying a relation between pretend play and executive functions in children 12-72 months.



All extracted effect sizes were entered into a **three-level random-effects meta-analysis** and explored for outliers, publication bias, and a number of moderators.



RESULTS

1 Main Meta-Analyses

	<i>k</i>	<i>r</i>	95% CI	<i>Q</i>	<i>F</i>
Main Analysis:	131	.17***	.13, .20	276.19***	
Outliers removed:	121	.16***	.13, .20	183.29***	

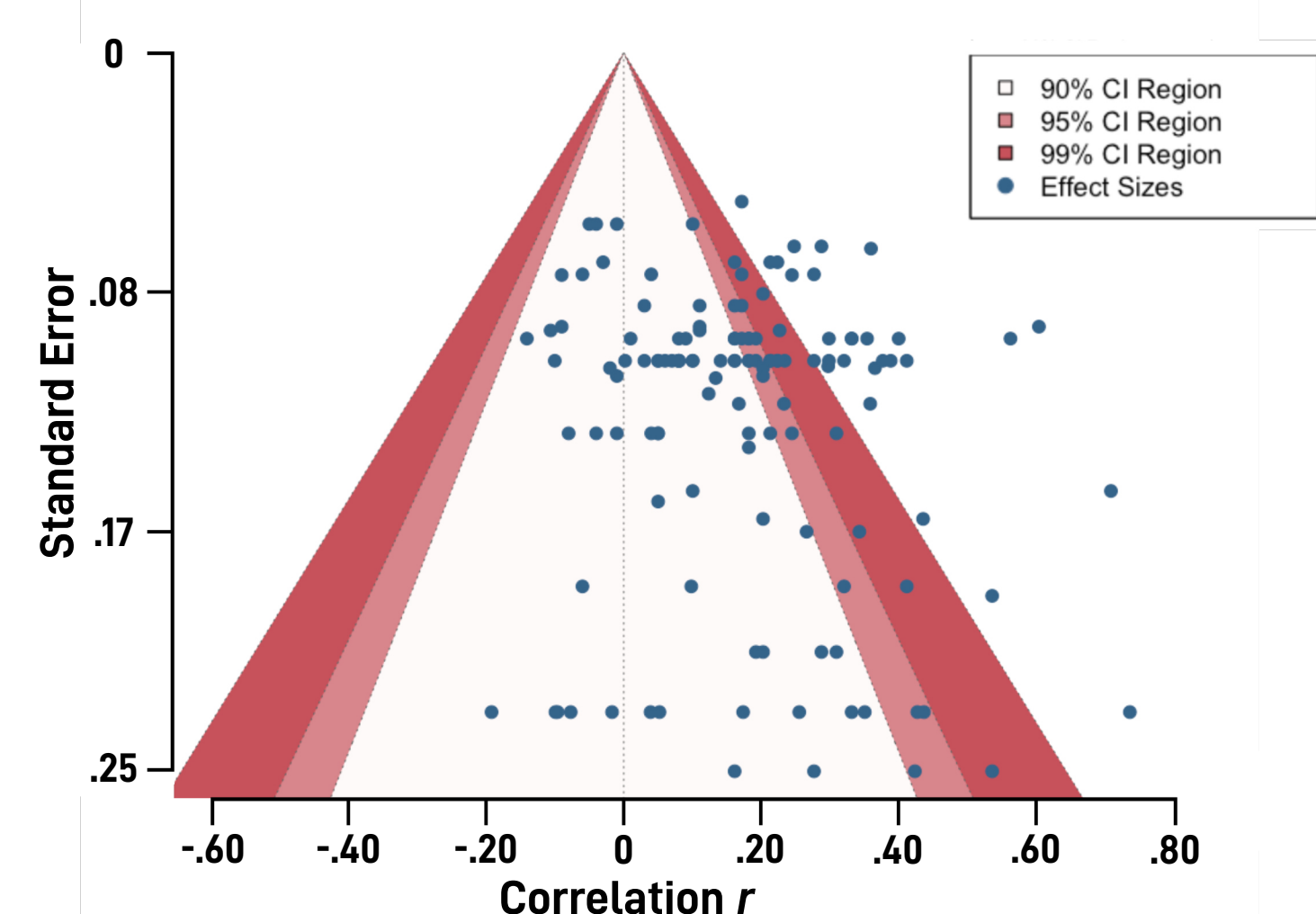
Moderator 1: Type of Pretend Play					
				210.94***	$F(1, 103) = 0.27$
Social Play:	46	.16***	.11, .21		
Solitary Play:	59	.18***	.14, .22		

Moderator 2: Executive Function Component					
				233.59***	$F(2, 120) = 0.10$
Inhib. Control:	71	.16***	.11, .20		
Cogn. Flexibility:	26	.17***	.11, .24		
Working Memory:	29	.15***	.09, .21		

this indicates that there's still variance unexplained.

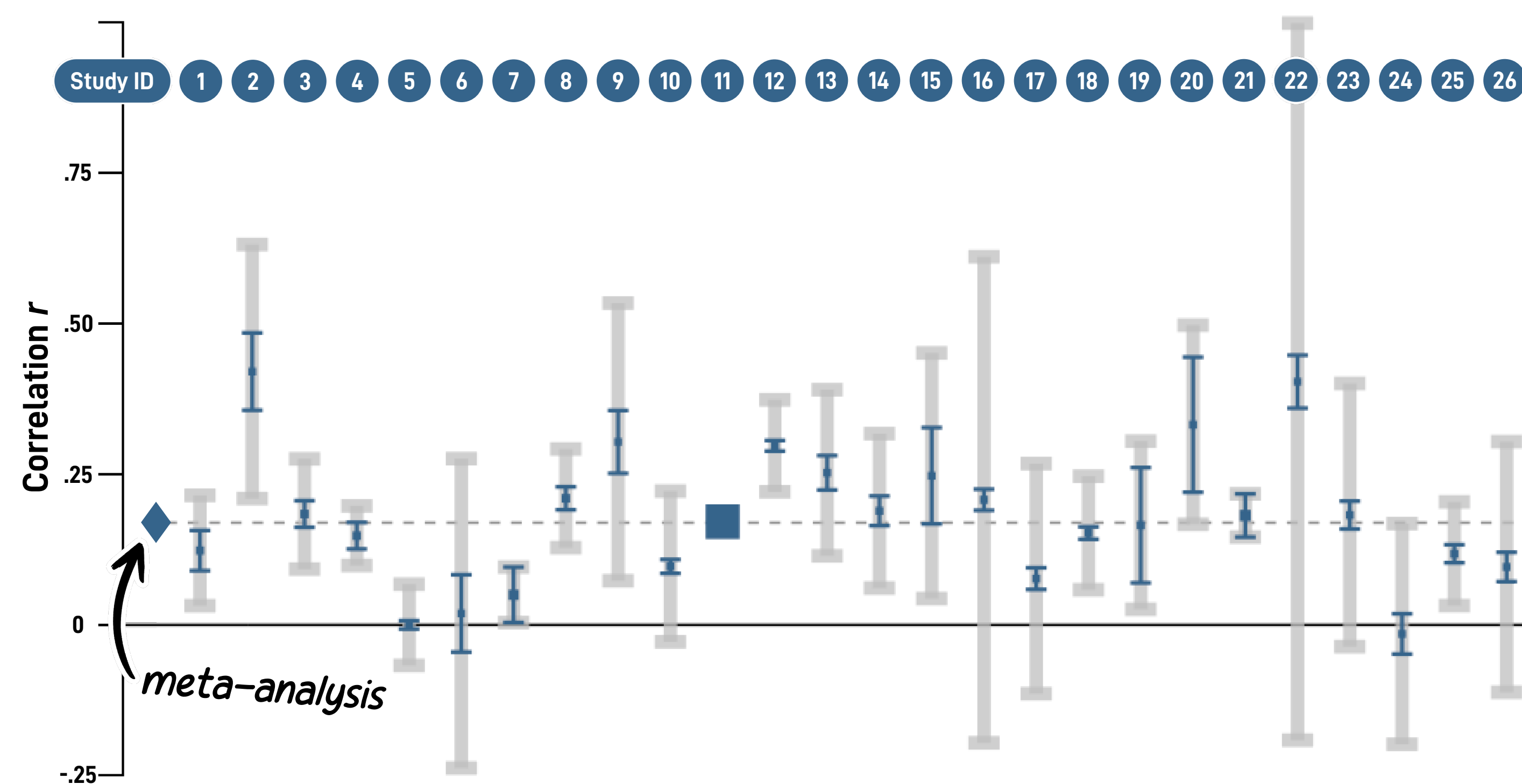
2 Publication Bias

A funnel plot, Egger's regression test, and the p-curve indicated no publication bias within the examined evidence.



3 Forest Plot

Enhanced forest plot of correlations in the 26 unique studies. Blue lines reflect the 95% confidence intervals that are based on the sample size of the study. Light gray lines are indicative of the study-level confidence-intervals and reflect the study precision.



CONCLUSIONS

This is the first quantitative meta-analysis on the relation of children's pretend play and their executive functions development.

- These findings **identify pretend play as a fertile context** within which to foster and encourage children's executive development in support of playful learning approaches in early education and intervention.
- There is a need for **further study to extend the current evidence base** to earlier periods of development (e.g., toddlerhood) and **study contextual factors** impacting children's play (e.g., parental playfulness).



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